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CAPSTONE PROJECT WORKSHOP #1

CAPSTONE PROJECT PLANNING & TRACKING

SEP490-ISP490 PROJECTS, FALL-2026

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Project Overview

- **Target**: students to work in teams of 5 students (4 or 6 students/team in some special cases), to complete a specific software application or system.
- **Product & scope**: as agreed with the mentor
- **Product size**: required to be equivalent to 20-25 medium UCs (3-7 transactions for each UC; action buttons on the screen, end user interactions to the software screens/functions or database transactions can be counted as transactions)
- **Dev Process**: full life cycle - req, design, code, test
- **Constraints:**
  - Time: 14 weeks (project reports 1-7)
  - Scope: >=75% of product scope completed
- **Templates/Guides:** https://cmshn.fpt.edu.vn/mod/folder/view.php?id=150581

Project Objectives

#	Metric	Unit	Planned	Actual	Notes / References
1	Effort Usage	Person-day			
2	Review Defects	No of defects			
3	Unit Test Defects	No of defects			
4	Integration Test Defects	No of defects			
5	System Test Defects	No of defects			
6	Acceptance Test Defects	No of defects			
7	Timeliness	%			Total on-time deliverables / Total project deliverables
8	Requirement Completeness	%			Failed if <=75%
9	…				

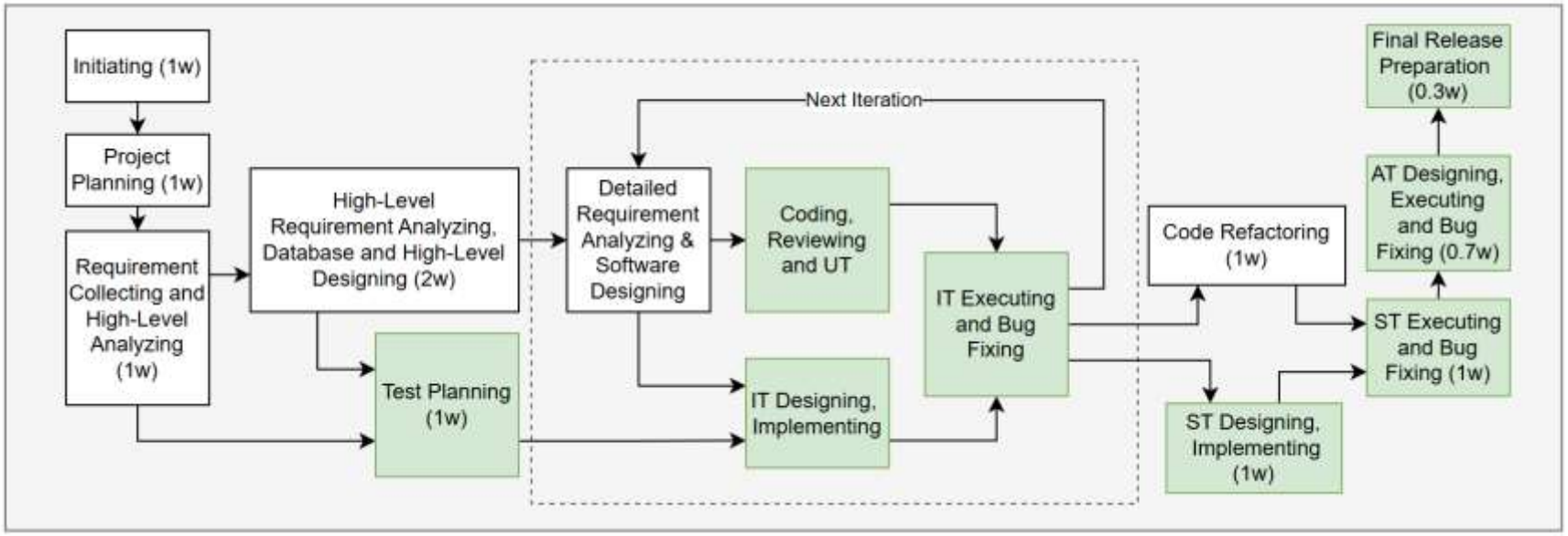
Defect rate: 4.2-9.0-13.4 defects/man-months (40-45-50% by reviewing)
Leakage rate: 0-0.13-0.53 defects/man-months

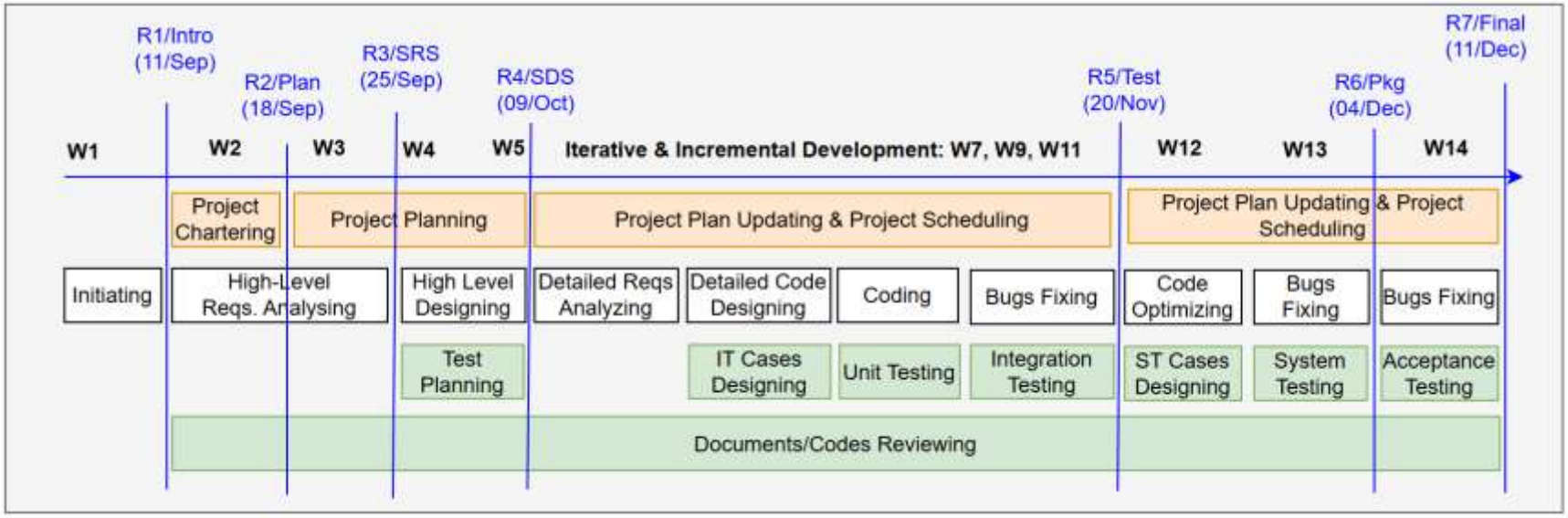
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Risks & Mitigations

#	Risk	Prob	Impact	Mitigation
R01	Requirements change significantly after Phase I Gate	Medium	High	Freeze SRS after Week 5. Change requests require impact analysis before acceptance. Scope change → negotiate timeline with PO.
R02	Coverage target below threshold at end of Iteration 2	Medium	High	Check JaCoCo every even week. If overall < [55%] after Iteration 2, allocate 1 dedicated day at start of Iteration 3 for test catch-up.
R03	Complex feature ([file upload / async job / state machine]) has hard-to-reproduce bugs	Medium	Medium	D1 writes unit tests for this feature with real file bytes (not mocks) from Week [N_1+3]. QA tests on staging (real DB) in Phase 3, not H2.
[R0x]	[Add project-specific risk]	[P]	[I]	[Mitigation]

High Level Schedule 1/5





High Level Schedule 2/5

#	Work Package	Est. Effort (pds)	Deadline
1	***Stage 1: Project Initiating (week 1)***		
1.1	Report 1 (Project Introduction)		Week1
2.	***Stage 2: Project Planning & Initial Requirements (weeks 2-3)***		
2.1	Report 3: SRS v0.9 (Overall Requirement)		Week2
2.2	Vertical Prototypes (Code Demo)		Week3
2.3	Report 2: Project Management Plan v1.0		Week3
3	***Stage 3: Software Design (weeks 4-5)***		
3.1	Vertical Prototypes (POC, or Proof Of Concepts)		Week4
3.2	Report 4: SDS v1.0 (High Level Design Specifications)		Week4
3.3	Report 5: Test Planning (ref Report5_Test Documentation)		Week5
3.4	Report 3: SRS v1.0 (Detailed Requirements for Iteration 1)		Week5
4	***Stage 4: Implementation (weeks 6-11)*** (Ref: Project Tracking, WBS sheet)		

High Level Schedule 3/5

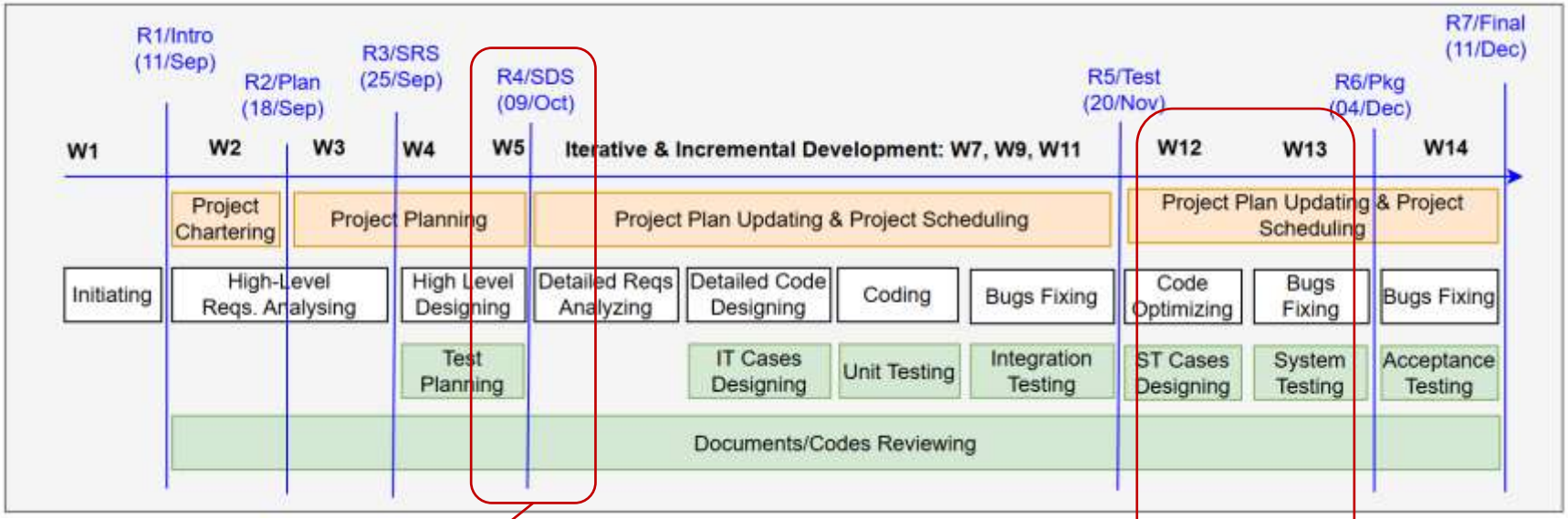
4	***Stage 4: Implementation (weeks 6-11)*** (Ref: Project Tracking, WBS sheet)		
4.1	Implementation Iteration 1 (weeks 6-7)		
4.1.1	Report 4: Detailed Design Specifications (SDS) v1.1		Week6
4.1.2	Code Package & Unit Testing Report		Week6
4. 1.3	Report 5: Integration Test Cases		Week6
4. 1.4	Report 5: Integration Test Report		Week7
4.1.5	Report 3: SRS v1.1 (Detailed Requirements for Iteration 2)		Week7
4.1.6	Software Package Version 1		Week7
4.2	Implementation Iteration 2 (weeks 8-9)		
4.2.1	Report 4: Detailed Design Specifications (SDS) v1.2		Week8
4.2.2	Code Package & Unit Testing Report		Week8
4. 2.3	Report 5: Integration Test Cases		Week8
4. 2.4	Report 5: Integration Test Report		Week9
4.2.5	Report 3: SRS v1.2 (Detailed Requirements for Iteration 3)		Week9
4.2.6	Software Package Version 2		Week9

High Level Schedule 4/5

4.3	Implementation Iteration 3 (weeks 10-11)		
4.3.1	Report 4: Detailed Design Specifications (SDS) v1.3		Week10
4.3.2	Code Package & Unit Testing Report		Week10
4.3.3	Report 5: Integration Test Cases		Week10
4.3.4	Report 5: Integration Test Report		Week11
4.3.5	Report 5: System Test Cases		Week11
4.3.6	Software Package Version 3		Week11
5	***Stage 5: Verification & Validation (weeks 12-13)***		
5.1	Report 5: System Test Report		Week12
5.2	Report 6: User Guides (Installation Guides, User Manuals)		Week12
5.3	Acceptance Test and Support		Week13
5.4	Final Software Package (Documents, Source Codes, etc.)		Week13
6	***Stage 6: Closing (week 14)***		
6.1	Report 7: Final Project Report		Week14
6.2	Presentation File		Week14

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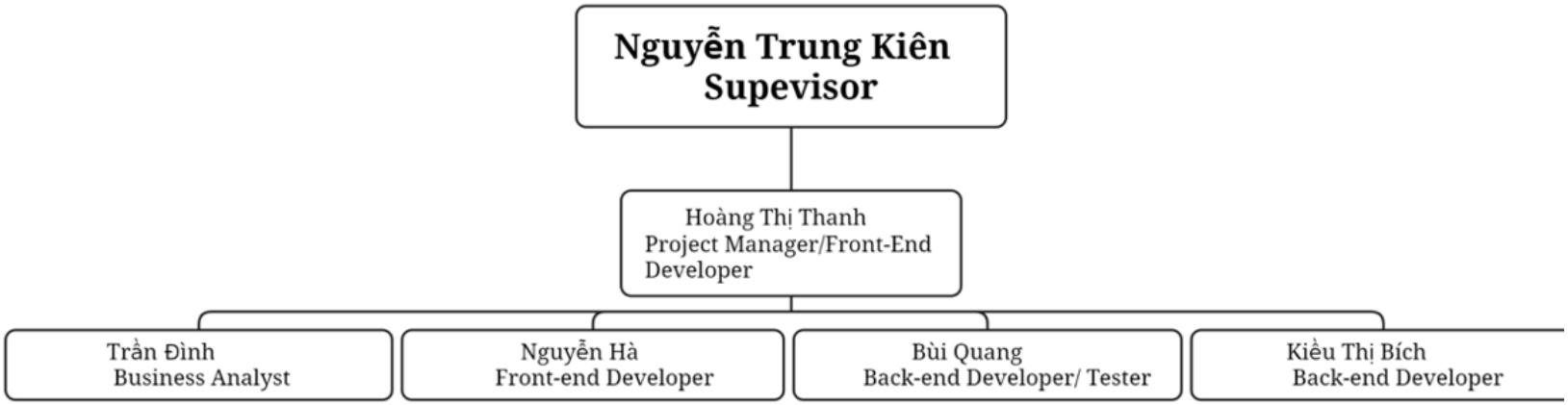
High Level Schedule 5/5



External Review 1.0:
Topic & Scope

External Review 1.1:
Project Products & Readiness for the final presentation

Responsibility Assignments



RACI Chart: R~Responsible, A~Accountable, C~Consulted, I~Informed

Work Package	ThanhHTHE01 4079	DinhTHE03580	HaNHE03235	QuangBHE048 71	BichKTHE0427
Work Package Name1	R	A	C	I	R
Work Package Name2	I	A	R	R	C
…					

Project Tracking
Outputs/Deliverables

	A	B	C	D	E	F	G
1	#	Work Package	Est. Effort (pds)	In Charge	Deadline	Status	Notes
2	***1***	***Stage 1: Project Initiating (week 1)***					
3	1.1	Report 1 (Project Introduction)			Week1		
4	***2***	***Stage 2: Project Planning & Initial Requirements (weeks 2-3)***					
5	2.1	Report 3: SRS v0.9 (Overall Requirement)			Week2		
6	2.2	Vertical Prototypes (Code Demo)			Week3		
7	2.3	Report 2: Project Management Plan v1.0			Week3		
8	***3***	***Stage 3: Software Design (weeks 4-5)***					
9	3.1	Vertical Prototypes (POC, or Proof Of Concepts)			Week4		
10	3.2	Report 4: SDS v1.0 (High Level Design Specifications)			Week4		
11	3.3	Report 5: Test Planning (ref Report5_Test Documentation)			Week5		
12	3.4	Report 3: SRS v1.0 (Detailed Requirements for Iteration 1)			Week5		
13	***4***	***Stage 4: Implementation (weeks 6-11)***					
35	***5***	***Stage 5: Verification & Validation (weeks 12-13)***					
36	5.1	Report 5: System Test Report			Week12		
37	5.2	Report 6: User Guides (Installation Guides, User Manuals)			Week12		
38	5.3	Acceptance Test and Support			Week13		
39	5.4	Final Software Package (Documents, Source Codes, etc.)			Week13		
40	***6***	***Stage 6: Closing (week 14)***					
41	6.1	Report 7: Final Project Report			Week14		
42	6.2	Presentation File			Week14		

Kiểm soát thay đổi 1/2

- Sử dung tính năng theo dõi của Ms Office, Google Docs/Sheets
- Lưu trữ lịch sử thay đổi (Records of Change)

Kiểm soát thay đổi 2/2

- Chèn thêm các đuôi phù hợp vào tên file
  - _v0.1, _v0.2, .., _v1.0, ..
  - _Version 1.1 draft 1,...
  - _220716.
- Sử dụng Git để quản lý Source Codes

Project Tracking

Product Requirements

	A	B	C	D	E	F	G	H	I	J
1	Function/Screen	Feature	Sub Feature	Function/Screen Description	Level*	Est. Effort	Planned	Status	Notes	
2	Home Page	Common			Medium	3	Iteration 2			
3	User Login	Common			Simple	4	Iteration 1			
4	User Register	Common			Simple	5	Iteration 1			
5	Reset Password	Common			Medium	3	Iteration 3			
6	User Authorization	Common			Complex	6	Iteration 1			
7	User Profile	Common			Simple	5	Iteration 3			
8	Change Password	Common			Simple	6	Iteration 3			
9	Blogs List	Public			Medium	3	Iteration 2			
10	Blog Details	Public			Simple	3	Iteration 2			
11	Posts List	Marketing			Medium	4	Iteration 1			
12	Post Details	Marketing			Medium	3	Iteration 1			
13	Users List	System			Simple	2	Iteration 1			
14	User Details	System			Simple	5	Iteration 1			

(*) You can rate the functions' complexity level based on the number of fields on the screens or the number of transactions in the function, with the details as below

Simple	<=7 fields	<=3 transactions
Medium	<=15 fields	<= 7 transactions
Complex	>15 fields	> 7 transactions

In which:

fields: screen components or database table fields

transactions can be counted actioning UI components, user or database access transactions

Planned	Functions	Total Effort
Iteration 1	7	29
Iteration 2	3	9
Iteration 3	3	14
Grand Total	**13**	**52**

Status: Pending, Analyzing, Analyzed, Coding, Coded, IT-Doing, IT-Done, ST-Doing, ST-Done, Cancelled

Daily Project Tracking

	A	B	C	D	E	F	G	H	I	J	
1	Function/Screen	Feature	Sub Feature	Role1	Role2	..	Screen Description	Level*	Planned	Status	Notes
2	Home Page	Common						Medium	Iteration 2	Analyzing	
3	User Login	Common						Simple	Iteration 1	Coded	
4	User Register	Common						Simple	Iteration 1	Pending	
5	Reset Password	Common						Medium	Iteration 3	Pending	
6	User Authorization	Common						Complex	Iteration 1	Pending	
7	User Profile	Common						Simple	Iteration 3	Pending	
8	Change Password	Common						Simple	Iteration 3	Pending	
9	Blogs List	Public						Medium	Iteration 2	Pending	
10	Blog Details	Public						Simple	Iteration 2	Pending	
11	Posts List	Marketing						Medium	Iteration 1	Pending	
12	Post Details	Marketing						Medium	Iteration 1	Pending	
13	Users List	System						Simple	Iteration 1	Pending	
14	User Details	System Admin						Simple	Iteration 1	Pending	

(*) You can rate the functions' complexity level based on the number of fields on the screens or the number of transactions in the function, with the details as below

Simple	<=7 fields	<=3 transactions
Medium	<=15 fields	<= 7 transactions
Complex	>15 fields	> 7 transactions

In which:

fields: screen components or database table fields

transactions can be counted actioning UI components, user or database access transactions

J	K	L	M	N	O	P	Q	R	S	T	
Status	SRS	SRS Status	IT Cases	IT Status	SDS	SDS Status	BE	BE Status	FE	FE Status	Notes
Analyzing	KienNT	To Do	KienNT	TC Done	KienNT		KienNT	Done	KienNT	Doing	
Coded	TuanTA	Doing	TuanTA	Test Done	TuanTA		TuanTA	Doing	TuanTA	To Do	
Pending											
Pending											

Student Evaluation

- On going evaluation (***50% of student GPA***, other 50% is by the thesis presentation council in the final presentation)

Evaluation Item	Weight
Project Idea & Introduction (Report 1)	*4.0%*
Project Management Plan (Report 2)	*8.0%*
Software Requirement (Report 3)	*16.0%*
Software Design (Report 4)	*18.0%*
Testing (Report 5)	*18.0%*
User Guides (Report 6)	*4.0%*
Implementation (Report 7)	*32.0%*

Who: supervisor/mentor/GVHD
When:
- After each report submission
- Might be updated right after report 7 submission

- Comments for thesis: mentors to be requested to send in the week 13

1. Tên Khóa luận tốt nghiệp/ Thesis title:

2. Họ tên những sinh viên bảo vệ khóa luận/ Students of the thesis defense:

3. Nhận xét của giảng viên hướng dẫn/ Comment from proposed Supervisor:

3.1 – Nội dung khóa luận (so với mục tiêu nghiên cứu, cơ sở lý luận, số liệu, phân tích, tính ứng dụng)/ Thesis content (compare to the research objectives, theoretical basis, data, analysis, application, etc)

3.2- Hình thức khóa luận (bố cục, phương pháp trình bày, tiếng Anh, trích dẫn)/ Thesis form (Layout, presentation methods, English, citation):

3.3- Thái độ của sinh viên trong quá trình làm khóa luận (tinh thần, thái độ của cả nhóm và từng thành viên trong nhóm, vai trò và đóng góp của từng thành viên trong nhóm/ Students' attitude (attitude of the group and each team member, roles and contributions of each team member)

4. Kết luận: Đạt ở mức nào? (hoặc không đạt)/ Conclusion: Pass at what stage? (Or not)

4.1. Mức độ đạt được so với mục tiêu (so với đề cương)/ Achievement level compare to the target (compare to the plan)

Hoàn thành mục tiêu đặt ra

4.2.Hạn chế/ Limitation .....

4.3. Ý kiến của giảng viên/ Proposed Supervisor comment (Đồng ý cho bảo vệ, Sửa lại để bảo vệ lần 2, Không được bảo vệ)

	Nguyễn Minh Hiếu *Student 1*	Đào Thiện Tuấn *Student 2*	Phạm Tuấn Tùng *Student 3*	Phạm Duy Thịnh *Student 4*
Kết luận/ *Conclusion*	■ Đồng ý cho bảo vệ/ *Agree to defense* □ Sửa lại để bảo vệ lần 2, nội dung	■ Đồng ý cho bảo vệ/ *Agree to defense* □ Sửa lại để bảo vệ lần 2, nội dung cần sửa/ *Revised for the 2nd defense,*	■ Đồng ý cho bảo vệ/ *Agree to defense* □ Sửa lại để bảo vệ lần 2, nội dung	■ Đồng ý cho bảo vệ/ *Agree to defense* □ Sửa lại để bảo vệ lần 2, nội

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Project Outputs/Deliverables

Week	Evaluation Items	Notes
W1	**Report 1_Project Introduction**	Project Idea & Researching Results
W2	**Report 2_Project Plan v0.9**	Scope/Purpose, Assumption Constraints, Capacity Planning, Risks, Management Approach
W3	**Report 3_SRS (Requirement)** *Report 2_Project Plan v1.0*	**High Level Requirements** *Updated Project Plan + Project Schedule*
W5	**Report 4_SDS (Design)** *Report 2_Project Plan v1.1* *Report 5_Test Plan (Word)*	**High Level Software Design, DB Design** *Updated Project Plan, Schedule, Tracking*
W11	**Report 5_Testing Documents** *Report 2_Project Plan v1.2*	**Test Plan, Test Cases/Reports, SW Package** *Updated Project Plan, Schedule, Tracking*
W13	**Report 6_User Guides** *Report 5_Testing Documents*	**Installation Guide, User Manual, SW Package** *Updated testing documents + Add ST Reports*
W14	**Report 7_Final Report** *Report 2_Project Plan v1.3* *Report 5_Testing Documents*	**Final Project Report/Package** *Final Project Plan and Tracking* *Updated testing documents, add AT Reports*

No thesis defense allowed for the teams who complete less than 75% of UCs, functions, or screens as submitted (as mentioned in the submitted Report 3) OR there are less than 20 average (3-7 transactions) use cases completed.


Q
A